

Item	Item Description	Units	y	Unit Cost	Total Cost
1	Mobilization and demobilization of Drilling rig and equipment to drill site	Item	1	\$ 2,500.00	\$ 2,500.00
2	Installation of surface casing.	M	4	\$ 420.00	\$ 1,680.00
3	Drilling of borehole of minimum diameter of 12" first 100 meter of hole 0-100m.	M	100	\$ 214.00	\$ 21,400.00
4	Drilling of borehole of minimum diameter of 12" second 100m of the hole (101m170m).	M	100	\$ 234.00	\$ 23,400.00
6	minimum diameter of 8" inch 16 bar of UPVC with bottom plug.	M	100	\$ 70.00	\$ 7,000.00
7	minimum diameter of 8" 1mm slot of 16 bar UPVC.	M	100	\$ 70.00	\$ 7,000.00
8	material.	Cum	8	\$ 180.00	\$ 1,440.00
9	Installation of gravel pack, seals and backfillings	Item	1	\$ 1,500.00	\$ 1,500.00
11	developing of the well till the water is clear	Ls	1	\$ 2,100.00	\$ 2,100.00
13	Install concrete reinforced slab on the well head	Ls	1	\$ 340.00	\$ 340.00
14	Install pump, riser pipes and connect electric	Ls	1	\$ 2,650.00	\$ 2,650.00
	TOTAL FOR BORHOLE DRILLING				\$ 71,010.00

BoQ of Elevated Water Tank for 25m³-					
	Description	Unit	QTY	Unit Rate	Total (USD)
ELEVATED TANK 25m³					
Sub-					
1	Mobilization and Clear site from vegetation, debris, unwanted material and dispose it to a	M³	20	\$ 10.00	\$ 200.00
2	Excavate 12 (2.0x2.0x1.5 deep)m holes for footings, level and compact well (before receiving foundation blinding) should be	M³	12	\$ 21.00	\$ 252.00
6	Lay 50mm thick lean concrete under footings 12(2.0x2.0x0.05)m concrete 1:4:8 /3bags of cem/0.4cum of river sand /0.8cum of Graded stone 40mm nominal size AS PER SPECS above the excavated footing and foundation trenches, cure the blinding	M³	1.1	\$ 110.00	\$ 121.00
3	Supply and backfill 700mm 12 (2.0x2.0x0.7) (imported or Excavated selected material over the ground floor compact as per Specification	M³	7.5	\$ 20.00	\$ 150.00
4	Provide and cast 12 (2.0x2.0x0.8)m R.C.C footing over the blinding mix 1:2:4 /6bags of cem/1cum of concrete Reinforced with double layered Y14 Bars spacing 150 mm C/C deformed steel in both directions/use vibrator	M³	4.3	\$ 400.00	\$ 1,720.00
5	Excavate masonry wall foundation (71.5mx0.4x0.5)m foundation, level and compact well (before receiving foundation	M³	7.5	\$ 70.00	\$ 525.00

6	Lay 50mm thick lean (71.5x0.05x0.45)m concrete 1:4:8 /3bags of cem/0.4cum of river sand /0.8cum of Graded stone 40mm nominal size AS PER SPECS above the excavated footing and foundation trenches, cure the	M ³	1.43	\$ 120.00	\$ 171.60
7	Masanory foundation wall : Construct (71.5x0.4x0.45)m foundation, level and compact well (before receiving foundation blinding)/cut and trim all projected faces	M ³	12.1	\$ 65.00	\$ 786.50
8	Provide and cast for Foundation R.C.C Coloumns 12(0.5x0.45x1.2)m. MIX 1:2:4 Reinforeced with 8Y16 all columns and 8Y20 with the center column linked with Y8mm	M ³	1.1	\$ 400.00	\$ 440.00
9	Provide and cast for Ground beam 400mm thick R.C.C (71.5x0.4x0.4)m R.C.C beam MIX 1:2:4 Reinforeced with 6Y14 linked with	M ³	1.8	\$ 400.00	\$ 720.00
	Provide and cast ground slab of dimension (12.5*8.5*0.2) R.C.C beam MIX 1:2:4/6bags of cem/1cum of concrete__ Reinforeced with 6Y12 @300mm c/c	M ³	12.5	\$ 400.00	\$ 5,000.00
10	Sub-Total				\$ 10,086.10
11	Super-Structure				
12	Coloums provide and cast 12(0.4x0.5x7.3m) R.C.C. MIX 1:2:4 ratio with 8Y16 all columns and 6Y20 of center column linked all with Y8mm stirupps @180mm c/c	M ³	3.85	\$ 400.00	\$ 1,540.00
13	Lintel beams Cast & Provide 71.5m long, 0.35mm Thick R.C.C Beams with in Mix 1:2:4 Reinforeced with 6Y16 mm for each Beam with linked Y8 bars, the space of the staffs not more than 180mm should check	M ³	1.3	\$ 400.00	\$ 520.00
14	Coloums provide and cast 9(0.45x0.45x3.05m) R.C.C beams 1:2:4 Reinforeced with 8Y16 linked with Y8mm	M ³	3.56	\$ 400.00	\$ 1,424.00
17	Provide and cast top tie beam of (71.5*0.4*0.4) R.C.C beam MIX 1:2:4/6bags of cem/1cum of concrete__ Reinforeced with 8Y16 linked with Y8mm stirupps @150mm	M ³	1.75	\$ 400.00	\$ 700.00
16	Elevated slab of below RCC walls : provide and cast 300mm thick (9.5*13*0.3) R.C.C beam MIX 1:2:4/6bags of cem/1cum of concrete__ Reinforeced with Y12 linked with	M ³	1.7	\$ 400.00	\$ 680.00

15	RCC WALLS: provide and cast 300MM RCC Wall Dimension are (42*2.65*0.3). Use 1:2:4 mixing ration of concrete reinforced with Y12 linked with Y12mm	M³	3.9	\$ 400.00	\$ 1,560.00
18	Provide and cast cover slab on RCC walls of dimension (8.5*13.5*0.15) R.C.C beam MIX 1:2:4/6bags of cem/1cum of concrete___ Reinforced with Y12 linked with Y12mm @200mm/c	M³	1.76	\$ 400.00	\$ 704.00
19	Supply and provide water proof coat internal walls of the tank	M2	150	\$ 4.00	\$ 600.00
20	IRON LIDS:Provide (500mmx500mm) Roof access cover	ls	1	\$ 140.00	\$ 140.00
Sub total					\$ 7,868.00
Finishing					
21	Wall Plaster: apply No 2 coats of 20mm thick plastering Mix 1:4 to intenal external walls including seiving sand and even finish as per line in levels , curing as per Field Engineer	SQM	120	\$ 7.00	\$ 840.00
22	white washing & Emulsion paint: apply No2 coats of emulsion paint to all external of the	SQM	120	\$ 7.00	\$ 840.00
23	G.I pipes :Provide and install 2inch class B inlet pipe, outlet pipe ,over flow pipe and	ls	16	\$ 20.00	\$ 320.00
24	LADDER :provide and fix steel ladder for elevated Tank up to the hiegh of 12m attached to steel bar of all grading Beams with an angle	ls	1	\$ 300.00	\$ 300.00
25	visibility: Attache 1.5x2 attached sign to the elevated water tank showing project	ls	1	\$ 500.00	\$ 500.00
Sub-total					\$ 2,800.00
TOTAL FOR ELEVATED WATER TANK					\$ 20,754.10

Borehole solarisation

Bacaadweyn					
No.	Item Description	Unit	Quantity	Rate	Amount
1	Mobilization: Clear the site from the debris, commence site establishment including provision of all required tool and Labour	LS	1	\$ 500.00	\$ 500.00
2	Solar panel: Supply and install solar panels of monocrystalline type 400W Efficiency between 17-24%	Pcs	36	\$ 300.00	\$ 10,800.00
3	supply stainless steel Solar pump inverter 30kW (DC/AC inverter)	Pcs	1	\$ 1,850.00	\$ 1,850.00

4	Solar structure:Supply and install the solar support mounting structure 1.5m height 0.5m deep made of steel frame (square tube 80mm x 80mmx5mm) mounting poles, diagonal support poles (square tube 50mmx 50mmx5mm) welded to the 3 mounting poles in each series and purlins (square tubes 50mmx50mmx3mm) with gentle slopping as required. The works includes cost for welding, Comomer fuse box and accessory DC circuit	No	36	\$ 70.00	\$ 2,520.00
5	combiner fuse box, Dc cercuit breaker manual transfer switch	Pcs	1	\$ 1,300.00	\$ 1,300.00
6	Case water proof and Well probe sensor	Pcs	1	\$ 100.00	\$ 100.00
7	Manual Change Over switch 100A		1	\$ 120.00	\$ 120.00
8	Installation accessories (Surge protector, Lightning Arrestors and earth).	LS	1	\$ 300.00	\$ 300.00
9	Installation Service	LS	1	\$ 1,500.00	\$ 1,500.00
10	supply and install DC wire for special solar cable 1x10mm2	M	200	\$ 3.00	\$ 600.00
11	supply and install pvc insulated double core cable 3x16mm2	M	30	\$ 4.00	\$ 120.00
Sub Total of Solar System					\$ 19,710.00

Item-2 Solar fence 30mx40					
NO.	Description	unit	Quantity	Rate	Amount
1	Foundation				
1.1	site clearance and romoving of trees and other scattered materials around the solar panels and mobilization of labour	LS	1	\$ 300.00	\$ 300.00
1.2	Excavation of foundation trenches 500mm depth and 500mm width	m3	20	\$ 60.00	\$ 1,200.00
1.3	Concreting on 200mm wide and 400mm depth trench at the base of chainlink by using concrete mix ratio 1:2:4.	m3	6.4	\$ 200.00	\$ 1,280.00
2	Fencing				\$ -
2.1	Supply and post angle 5mm thick with 2000mm long each by 2 coat painting	m	80	\$ 15.00	\$ 1,200.00
2.2	Supply and install 180cm high standard steel galvanized mesh supported by 5 inch angular metalic bars 2300mm high installed at the ground level @2000mm spacing inluding Holding angles	m	140	\$ 13.00	\$ 1,820.00
2.3	supply and fix steel gate size of 4m wide and 2m high consisting of 2 equally sized leafs, completed with door locks, latches and 2 coat painting	Pcs	1	\$ 300.00	\$ 300.00
Sub Total of Fencing					\$ 6,100.00

	Total Cost of solar system and fencing				\$ 25,810.00
Pumping- of Borehole					
No.	Description	Unit	Quantit	Unit Prices	Total Cost
	Supply and intallation of new perkin Generator 45KVA three phase	PCS	1	\$ 15,000.00	\$15,000.00
2	Supply and install Lowara Summersible pump	PCS	1	\$ 5,500.00	\$5,500.00
3	Supply and intall control panel 25hp	PCS	1	\$ 550.00	\$550.00
4	Supply and conduct to summersible pump the Das GT-2 Glue for cable in water pump	PCS	2	\$ 45.00	\$90.00
5	provide and fix Pvc insulated double core Cable 4x16mm2	M	230	\$ 14.50	\$3,335.00
6	Supply and intall 3 inch PVC with all Neccesary Fittings.	No.	30	\$ 55.00	\$1,650.00
7	Supply and install all neccesary fattings such as top and bottom adapters, reducers, nipipes, Elbows and paint	LS	1	\$ 950.00	\$950.00
8	Installation of borehole Equipment by using	LS	1	\$ 2,550.00	\$2,550.00
	Sub Total of Pumping				\$29,625.00
GENERATOR ROOM					
M	DESCRIPTION	UNIT	TY	RATE (USD)	(USD)
	SUBSTRUCTURE (Provisional)				
	Excavation				
	supporting sides and keeping free from water, mud and fallen materials by bailing, pumping or otherwise				
A	Prepare site by stripping top 200 mm of soil to remove all debris including sand (if any) from site and carting away spoil	m ²	34	\$ 7.00	\$ 238.00
B	Excavate to reduce levels not exceeding 1.50m deep average depth 300mm	m ²	34	\$ 6.00	\$ 204.00
C	reduced levels depth not exceeding 1.50m deep	m ³	28	\$ 25.00	\$ 700.00
E	Remove surplus excavated material from site	m ³	12	\$ 25.00	\$ 300.00
F	Backfill around foundation	m ³	26	\$ 4.00	\$ 104.80
	Filing				\$ -
G	300 mm thick approved hardcore filling spread, well rammed and compacted in 150mm layers to receive concrete surface bed	m ³	5	\$ 11.00	\$ 55.00
	Insitu concrete: class 15: mix 1:3:6				\$ -
H	50mm blinding layer under foundations	m ²	10	\$ 5.00	\$ 50.00
I	50mm blinding layer on hardcore surfaces	m ²	15	\$ 4.00	\$ 60.00
J	insecticide	m ²	15	\$ 33.00	\$ 495.00
	Concrete work				\$ -
	Reinforced Concrete class 25				\$ -
K	Strip foundation	m ³	2	\$ 21.00	\$ 42.00
L	100mm thick floor slab with surface steel trowelled smooth	m ²	19	\$ 15.00	\$ 285.00
	Total page 1 to Substructures Collection				\$ -
	Reinforcement				\$ -
A	reinforcement bars	kg	55	\$ 3.00	\$ 165.00

B	Mesh fabric reinforcement ref. No. A142 laid in floor slab with minimum 150 mm side allowance	m ²	19	\$ 3.00	\$ 57.00
	Sawn formwork				\$ -
C	Formwork to edges of floor slab girth over 75mm but not exceeding 150mm	m	17	\$ 3.00	\$ 51.00
	Walling				\$ -
D	300mm Thick rubble stone foundation walling in cement and sand mortar (1:3)	m ²	21	\$ 35.00	\$ 735.00
E	One layer 1000gauge polythene sheet damp proof membrane under beds: 300mm laps	m ²	19	\$ 4.00	\$ 76.00
F	200mm wide Bituminous felt damp-proof course	m	17	\$ 4.00	\$ 68.00
	Plinths				\$ -
G	plinths	m ²	5	\$ 4.00	\$ 20.00
H	Prepare and apply three coats black bituminous paint to rendered plinths externally	m ²	5	\$ 5.00	\$ 25.00
	Total page 2 to Substructures Collection				\$ -
	<u>SUBSTRUCTURES COLLECTION</u>				\$ -
	Total from page 1				\$ -
	Total from page 2				\$ -
	SUBSTRUCTURES SUB TOTAL TO SUMMARY				\$ -
	SUPERSTRUCTURES				\$ -
	Walling				\$ -
A	walling	m ²	38	\$ 18.00	\$ 684.00
	<u>Wall Coping</u>				\$ -
B	<u>Precast concrete 600 x 300 x 50 mm Thick twice weathered and throated coping jointed and pointed in cement and sand mortar</u>	m	13	\$ 30.00	\$ 390.00
	Concrete Work				\$ -
	<u>Vibrated reinforced concrete class 25 (1:1.5:3) with 20mm maximum aggregate as described in:</u>				\$ -
A	Ring Beam	m ³	1	\$ 45.00	\$ 45.00
B	Columns	m ³	1	\$ 45.00	\$ 45.00
C	100mm Thick suspended seat /bench	m ²	4	\$ 45.00	\$ 180.00
	Reinforcement				\$ -
D	8mm Diameter high tensile reinforcement bar	kg	30	\$ 3.00	\$ 90.00
E	Ditto but 10mm	kg	55	\$ 3.00	\$ 165.00
F	Ditto but 12mm	Kg	25	\$ 3.00	\$ 75.00
G	Mesh fabric reinforcement ref. No. A142 laid in floor slab with minimum 150 mm side allowance	m ²	2	\$ 15.00	\$ 30.00
	Sawn Formwork to:				\$ -
H	Sides and soffits of ring beam	m ²	12	\$ 4.00	\$ 48.00
I	Vertical sides of columns	m ²	5	\$ 4.00	\$ 20.00
J	Sides and soffits of concrete seat	m ²	5	\$ 4.00	\$ 20.00
	Total page 3 to Superstructures Collection				\$ -
	Roofing				\$ -
	<u>Roof Structure</u>				\$ -
	<u>Sawn celcured cypress timber as described in:</u>				\$ -
A	200mm x 25mm Fascia Board	m	4	\$ 4.00	\$ 16.00
B	100x50mmRafters	m	14	\$ 4.00	\$ 56.00
C	100mm x 50mm wall plate	m	8	\$ 4.00	\$ 32.00
D	50 x 50 Purlins	m	21	\$ 12.00	\$ 252.00
	<u>Roof Covering</u>				\$ -

E	30 Gauge galvanized corrugated iron sheets fixed to timber Purlins	m ²	19	\$ 4.00	\$ 76.00
F	26 Gauge flushing 450mm wide bent to shape tucked under roofing sheets and parapet wall	m	13	\$ 4.00	\$ 52.00
	Painting and Decorating				\$ -
G	Knot prime stop and apply two undercoats and one gloss finishing coat oil paint to fascia board 200- 300 mm wide	m	4	\$ 11.00	\$ 44.00
	Total page 4 to Superstructures Collection				\$ -
	Doors and Windows				\$ -
A	45mm thick Match boarded timber single door, overall size 900x 2100mm high	No	1	\$ 230.00	\$ 230.00
	<u>Wrot cypress 1st grade</u>				\$ -
B	150x50mm frame two labours plugged screwed and pelltated	m	5	\$ 4.00	\$ 20.00
C	50x25mm architrave two labours	m	5	\$ 4.00	\$ 20.00
D	25x15mm quadrant one labour	m	5	\$ 4.00	\$ 20.00
	<u>Supply and fix the following ironmongery to timber with matching screws</u>				\$ -
E	Stainless steel hinges	Pairs	2	\$ 6.00	\$ 12.00
F	Three lever mortice lock	No	1	\$ 6.00	\$ 6.00
G	Two lever mortice lock (Optional)	No	1	\$ 6.00	\$ 6.00
H	Rubber door stop (Optional)	No	1	\$ 6.00	\$ 6.00
	Total page 5 to Superstructures Collection				\$ -
	<u>WINDOWS</u>				\$ -
	Supply and Fix the following purpose made steel Lourved windows manufactured from heavy duty standard sections primed with one coat of red oxide primer and fixing to masonry or concrete head and jambs complete with closing mechanism and other fittings.				\$ -
A	Window overall size 1000x900mm high	No	2	\$ 110.00	\$ 220.00
	<u>5mm thick clear sheet glass bedded in mastic</u>				\$ -
B	In panes of various sizes	m ²	3	\$ 25.00	\$ 75.00
	<u>windows; 10mm square section steel bars smooth welded together at 75mm centres both ways fixed with lugs built into walls; one coat shop primer</u>				\$ -
C	Grille overall size 1000x900mm high	No	3	\$ 25.00	\$ 75.00
D	Precast concrete window cill size 260 x 50mm Thick sunk - weathered and throated and bedded and jointed in cement sand mortar	m	3	\$ 20.00	\$ 60.00
E	Prepare and apply two undercoats and one finishing coat oil paint to timber door	m ²	4	\$ 5.00	\$ 20.00
F	Ditto windows	m ²	5	\$ 5.00	\$ 25.00
	Total page 6 to Superstructures Collection				\$ -
	Finishes				\$ -
	<u>Floor Finishes</u>				\$ -
	<u>Cement and sand mortar (1:3) in:</u>				\$ -
A	30mm thick steel trowelled screed	m ²	15	\$ 25.00	\$ 375.00
B	100 x 25mm Thick skirting to junction with floor and wall finish	m	13	\$ 5.00	\$ 65.00
	<u>Wall Finishes</u>				\$ -

	Cement and sand mortar (1:3) rendering in:				\$ -
C	12 mm Thick mortar rendering with steel float finish to masonry and concrete surfaces	m ²	38	\$ 5.00	\$ 190.00
	12mm lime plaster: steel trowelled finish: on concrete, block work or stonework: to				\$ -
D	Walls and beams	m ²	33	\$ 5.00	\$ 165.00
	paint to:				\$ -
E	Rendered surfaces	m ²	38	\$ 5.00	\$ 190.00
F	Plastered surfaces	m ²	33	\$ 5.00	\$ 165.00
	Total page 7 to Superstructure Collection				\$ -
	<u>SUPERSTRUCTURE COLLECTION</u>				\$ -
	Total from page 3				\$ -
	Total from page 4				\$ -
	Total from page 5				\$ -
	Total from page 6				\$ -
	Total from above 7				\$ -
	SUMMARY				\$ -
	GENERAROR ROOM				\$ -
	SUMMARY				\$ -
	(1) SUBSTRUCTURE				\$ -
	(2) SUPERSTUCTURE				\$ -
	Grand Total For Caretakers Room				\$ 7,995.80
Summary-of borehole drilling and borehole upgrading					
No.	Description	Unit	Quanti	Unit Prices	Total Cost
1	Borehole drilling	Nos	1	\$ 71,010.00	\$ 71,010.00
2	Total cost of elevated water tank	Nos	1	\$ 20,754.10	\$ 20,754.10
3	Total cost of Solar system installation and	Nos	1	\$ 25,810.00	\$ 25,810.00
4	Total cost of Pump and rising Pipe installation	Nos	1	\$ 29,625.00	\$ 29,625.00
5	Total cost of Generator room	Nos	1	\$ 7,995.80	\$ 7,995.80
	Grand Total For all infrastructure of borehole				\$155,194.90
	Grand Total For all infrastructure of 2 boreholes				